

## F-Strings:

1. Ask the user to enter their city name. Print: Welcome to {city}!

```
city = input("Enter your city name: ")  
print(f"Welcome to {city}!")
```

```
Enter your city name: kadapa  
Welcome to kadapa!
```

2. Ask the user to enter two numbers. Print their product using an f-string.

```
num1 = float(input("Enter first number: "))  
num2 = float(input("Enter second number: "))  
  
product = num1 * num2  
print(f"The product of {num1} and {num2} is {product}")
```

```
Enter first number: 1  
Enter second number: 2  
The product of 1.0 and 2.0 is 2.0
```

3. Ask the user for their name and age. Print: Hi {name}, you will be {age+1} next year!

```
name = input("Enter your name: ")  
age = int(input("Enter your age: "))  
  
print(f"Hi {name}, you will be {age + 1} next year!")
```

```
Enter your name: akila  
Enter your age: 20  
Hi akila, you will be 21 next year!
```

4. Ask the user to enter a temperature in Celsius. Convert it to Fahrenheit and print the result using an f-string.

Formula:  $F = C \times \frac{9}{5} + 32$

```
celsius = float(input("Enter temperature in Celsius: "))  
  
fahrenheit = celsius * 9/5 + 32  
  
print(f"{celsius}°C is equal to {fahrenheit}°F")
```

```
Enter temperature in Celsius: 35  
35.0°C is equal to 95.0°F
```

## For Loop:

### 1. Print your name 5 times using a for loop

```
for i in range(5):  
    print("Akila")
```

```
Akila  
Akila  
Akila  
Akila  
Akila
```

### 2. Print all numbers from 1 to 10

```
for i in range(1, 11):  
    print(i)
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```

### 3. Print the multiplication table of any number the user enters (1x to 10x)

```
num = int(input("Enter a number: "))  
  
for i in range(1, 11):  
    print(f"{num} x {i} = {num * i}")
```

```
Enter a number: 64  
64 x 1 = 64  
64 x 2 = 128  
64 x 3 = 192  
64 x 4 = 256  
64 x 5 = 320  
64 x 6 = 384  
64 x 7 = 448  
64 x 8 = 512  
64 x 9 = 576  
64 x 10 = 640
```

### 4. Print all odd numbers between 1 and 20

```
for i in range(1, 21, 2):  
    print(i)
```

```
1  
3  
5  
7  
9  
11  
13  
15  
17  
19
```

5. Print a countdown from 10 down to 1, then print "Blast off!"

```
for i in range(10, 0, -1):  
    print(i)
```

```
print("Blast off!")
```

```
10  
9  
8  
7  
6  
5  
4  
3  
2  
1  
Blast off!
```

6. Print the sum of all numbers from 1 to 100 using a for loop

```
total = 0
```

```
for i in range(1, 101):  
    total += i
```

```
print(f"The sum of numbers from 1 to 100 is {total}")
```

```
The sum of numbers from 1 to 100 is 5050
```

Lists:

1. Create a list of 5 favourite movies and print the full list

```
movies = ["Bahubali", "RRR", "Pushpa", "KGF", "Salaar"]

print("Favourite Movies:")
print(movies)
```

```
Favourite Movies:
['Bahubali', 'RRR', 'Pushpa', 'KGF', 'Salaar']
```

## 2. Print only the first and last movie using indexing

```
movies = ["Bahubali", "RRR", "Pushpa", "KGF", "Salaar"]

print("First Movie:", movies[0])
print("Last Movie:", movies[-1])
```

```
First Movie: Bahubali
Last Movie: Salaar
```

## 3. Add one more movie at the end using append(). Print the updated list.

```
movies = ["Bahubali", "RRR", "Pushpa", "KGF", "Salaar"]

movies.append("Devara")

print("Updated List:")
print(movies)
```

```
Updated List:
['Bahubali', 'RRR', 'Pushpa', 'KGF', 'Salaar', 'Devara']
```

## 4. Remove the second movie using remove(). Print the updated list.

```
movies = ["Bahubali", "RRR", "Pushpa", "KGF", "Salaar"]

movies.remove("RRR")

print("Updated List:")
print(movies)
```

```
Updated List:
['Bahubali', 'Pushpa', 'KGF', 'Salaar']
```

## 5. Sort the list alphabetically and print it.

```
movies = ["Bahubali", "RRR", "Pushpa", "KGF", "Salaar"]

movies.sort()

print("Sorted List:")
print(movies)
```

```
Sorted List:  
['Bahubali', 'KGF', 'Pushpa', 'RRR', 'Salaar']
```

6. Traverse the list with a for loop and print each movie on a new line.

```
movies = ["Bahubali", "RRR", "Pushpa", "KGF", "Salaar"]  
  
for movie in movies:  
    print(movie)
```

```
Bahubali  
RRR  
Pushpa  
KGF  
Salaar
```

7. Create a list of 5 numbers. Print the largest number using sort().

```
numbers = [45, 12, 78, 34, 56]  
  
numbers.sort()  
  
print("Largest Number:", numbers[-1])
```

```
Largest Number: 78
```

Tuples:

1. Create a tuple of 4 Indian state names and print it.

```
states = ("Andhra Pradesh", "Telangana", "Kerala", "Tamil Nadu")  
  
print(states)
```

```
('Andhra Pradesh', 'Telangana', 'Kerala', 'Tamil Nadu')
```

2. Print the second and last state using indexing.

```
states = ("Andhra Pradesh", "Telangana", "Kerala", "Tamil Nadu")  
  
print("Second State:", states[1])  
print("Last State:", states[-1])
```

```
Second State: Telangana  
Last State: Tamil Nadu
```

3. Loop through the tuple and print each state on a new line.

```
states = ("Andhra Pradesh", "Telangana", "Kerala", "Tamil Nadu")

for state in states:
    print(state)
```

```
Andhra Pradesh
Telangana
Kerala
Tamil Nadu
```

4. Count how many times 'Goa' appears in ('Kerala', 'Goa', 'Goa', 'Tamil Nadu')

```
states = ("Kerala", "Goa", "Goa", "Tamil Nadu")

count = states.count("Goa")

print("Goa appears", count, "times")
```

```
Goa appears 2 times
```

5. Find the index of 'Tamil Nadu' in the tuple above.

```
states = ("Kerala", "Goa", "Goa", "Tamil Nadu")

index = states.index("Tamil Nadu")

print("Index of Tamil Nadu:", index)
```

```
Index of Tamil Nadu: 3
```

6. Try changing a value in a tuple. Write down the error message you get.

```
states = ("Andhra Pradesh", "Telangana", "Kerala", "Tamil Nadu")

states[1] = "Goa"
```

```
-----
TypeError                                Traceback (most recent call last)
/tmp/ipykernel_7055/794840523.py in <cell line: 0>()
      1 states = ("Andhra Pradesh", "Telangana", "Kerala", "Tamil Nadu")
      2
----> 3 states[1] = "Goa"

TypeError: 'tuple' object does not support item assignment
```

Next steps: [Explain error](#)

TypeError: 'tuple' object does not support item assignment